

MMD II Project Report Topic 3, Team 1:

Multilingual Emotion Detection

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Abstract—We address multi-label emotion detection across six languages using Qwen3.5 [1], a multilingual decoder transformer, trained on English, Swahili, and Ukrainian from the BRIGHTER shared task [2]. Rather than attaching a classification head, we treat emotion detection as constrained JSON generation. A fine-tuned 0.8B model (requiring only 1.5 GB at inference) surpasses every larger base model on trained languages, including a model 11 times its size. However, fine-tuning at scale erodes zero-shot transfer: for languages unseen during training, a large unmodified base model is preferable. LIME [3] analysis reveals that word-level attributions transfer across scripts for high-resource languages but collapse for low-resource ones such as Yoruba.

I. INTRODUCTION

Automatic emotion recognition in user-generated text is increasingly important for applications ranging from mental-health monitoring to social-media analysis. The BRIGHTER shared task (SemEval-2025 Task 11) [2] poses this as a multilingual, multi-label problem: given a short text in any of 28 languages, predict which subset of six emotions (anger, disgust, fear, joy, sadness, surprise) it expresses. Three challenges make this non-trivial: (i) the output is multi-label, (ii) the label distribution is heavily skewed (English has no disgust annotations in the training data at all), and (iii) the languages span radically different scripts, resource levels, and morphological families. We select English, Swahili, and Ukrainian for training, representing high-resource Indo-European, Bantu, and East Slavic families, and evaluate zero-shot transfer to German, Brazilian Portuguese, and Yoruba.

Rather than fine-tuning a discriminative encoder such as mBERT [4] (which we run as a baseline), we treat classification as constrained text generation using Qwen3.5 [1], a multilingual decoder transformer. The model emits a strict JSON object enumerating the active emotion labels; this single output contract is language-agnostic and handles any label subset natively without a per-task classification head, and the same generation recipe scales across model sizes (0.8B through 9B) without architectural changes.

Our results reveal a practical tension between two deployment objectives. A fine-tuned Qwen3.5-0.8B (requiring only 1.5 GB at inference, small enough for on-device use on modern smartphones) surpasses every larger base model on trained languages, including an $11\times$ larger 9B variant, because task-specific fine-tuning compensates for smaller capacity. However, fine-tuning at scale erodes zero-shot transfer: for languages unseen during training, a larger base model without fine-tuning is the better choice.

II. METHODOLOGY

A. Model and Task Framing

We use Qwen3.5 [1], a multilingual decoder transformer, as our primary model, with the 0.8B variant as the primary deliverable and a scaling study covering 2B and 4B (full fine-tune) and 9B (base only). Classification is posed as structured generation: a terse system prompt lists the six emotion labels and the expected JSON schema, while the user turn provides the language name and text to classify, the user prompt then asks the model to generate the classification JSON in its turn. The model generates the format {"emotions": ["predicted label1", "predicted label2", ...]}, where an empty list signals no emotion. For the 4B and 9B variants, which are reasoning models that would otherwise emit chain-of-thought before any answer, we disable thinking at generation time to preserve structured JSON output and keep inference costs comparable across sizes.

B. Data and Split Protocol

We use the BRIGHTER dataset [2] (SemEval-2025 Task 11). Training languages provide 8,537 examples in total: English (2,764), Swahili (3,307), and Ukrainian (2,466). The official training split is partitioned 85/15 into internal train and internal development sets using seed 42069; the official development split is withheld and used exactly once for final evaluation. Training loss is computed only on the JSON completion tokens, prompt tokens are masked, so the gradient signal is concentrated on label prediction rather than prompt memorisation.

C. Training Setup

A learning-rate, epoch, and weight-decay sweep¹ on the internal development set converged to $\text{lr} = 10^{-5}$, 3 epochs, a cosine schedule with 3% linear warmup, and weight decay 0.01. All other tested configurations landed within the multi-seed noise floor (± 0.008 macro-F1, $n = 5$ seeds), confirming no remaining hyperparameter headroom. Effective batch size is 16 (4 per device $\times$ 4 gradient-accumulation steps). Full fine-tuning of 2B and 4B uses 8-bit AdamW (bitsandbytes) [5] to fit within a 24 GB GPU; 9B full fine-tuning requires ~ 144 GB of optimizer state and is therefore evaluated at base only.

¹Grid search on internal development set: learning rates 5×10^{-6} , 7×10^{-6} , 10^{-5} , 2×10^{-5} , 3×10^{-5} ; epochs 1–4 (macro-F1 evaluated after each epoch); weight decay 0.01–0.1.

D. Evaluation

The model’s generated JSON is parsed into a 6-dimensional binary label vector; failed parses yield empty predictions. The primary metric is macro-F1 over the six labels. Macro averaging is preferred over micro-F1 because the label distribution is heavily imbalanced, and macro-F1 penalises a model that ignores any single label. Per-label F1 is additionally reported to identify the hardest emotions to predict. The untrained base model is scored on the same development samples as its fine-tuned counterpart, so $\Delta = \text{fine-tuned} - \text{base}$ isolates the contribution of supervised fine-tuning. LIME [3] explanations are generated for one selected example per language, treating discrete generation outputs as proxy probabilities.

III. RESULTS

A. Baseline: mBERT

We fine-tuned mBERT [4], a 110M-parameter multilingual encoder, with six sigmoid output heads and binary cross-entropy loss on the same training split. On the official development set, mBERT achieves a trained-language macro-F1 mean of 0.393 and a zero-shot mean of 0.248. Notably, mBERT outperforms the fine-tuned Qwen3.5-0.8B on Yoruba (0.169 vs. 0.097): the encoder’s sigmoid outputs are less biased toward abstention than Qwen’s generation, which rarely emits positive labels for scripts underrepresented in its pretraining data. On all other languages, the generative approach is substantially stronger.

B. Qwen3.5-0.8B Fine-Tuning

TABLE I

OFFICIAL-DEV MACRO-F1. BASE = UNTRAINED QWEN3.5-0.8B; FT = FINE-TUNED. BOLD MARKS THE BEST RESULT PER LANGUAGE. $\Delta = \text{FT} - \text{BASE}$.

Language	Status	mBERT	Base	FT 0.8B	Δ
English	trained	0.541	0.287	0.600	+0.313
Swahili	trained	0.238	0.176	0.273	+0.097
Ukrainian	trained	0.400	0.300	0.501	+0.201
German	zero-shot	0.310	0.302	0.391	+0.089
Brazilian Port.	zero-shot	0.266	0.176	0.396	+0.220
Yoruba	zero-shot	0.169	0.045	0.097	+0.052
Mean (trained)		0.393	0.254	0.458	+0.204
Mean (zero-shot)		0.248	0.174	0.295	+0.120

Fine-tuning the 0.8B base model on the three training languages yields a trained-language macro-F1 of 0.458, a +0.204 absolute gain over the untrained base (0.254). Fine-tuning also improves zero-shot languages by +0.120 mean, confirming the recipe teaches genuine emotion knowledge beyond prompt formatting. The hardest label is disgust, which scores 0.00 for English due to zero positive training examples. Swahili fear (0.100) is the most volatile cell across seeds (0.068 ± 0.070 , $\text{std} \approx \text{mean}$, effectively a coin flip).

C. Model Scaling Study

TABLE II

OFFICIAL-DEV MACRO-F1 BY MODEL SIZE (TR. = TRAINED LANGUAGES; ZS. = ZERO-SHOT LANGUAGES; VRAM AT BF16 INFERENCE). 9B FT IS INFEASIBLE ON A 24 GB GPU. BOLD MARKS THE BEST RESULT PER LANGUAGE.

Size	Base tr.	Base zs.	FT tr.	FT zs.	VRAM
0.8B	0.255	0.174	0.468	0.295	1.5 GB
2B	0.323	0.311	0.495	0.328	3.8 GB
4B	0.410	0.395	0.543	0.362	8.5 GB
9B	0.431	0.405	n/a	n/a	17.9 GB

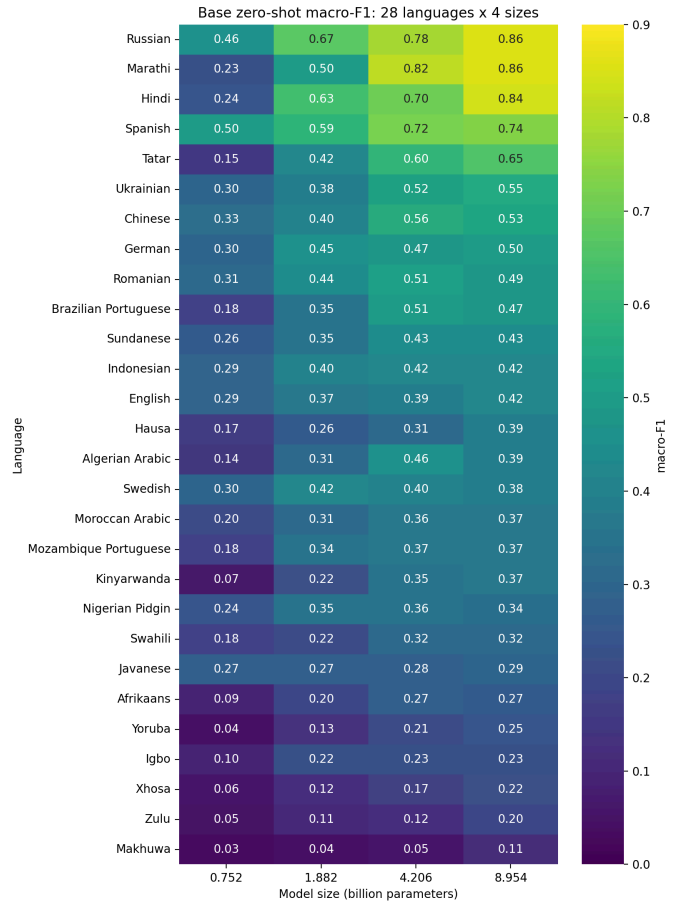


Fig. 1. Per-language macro-F1 across model sizes (base models, official dev). Rows are sorted by 9B score.

Two headline findings emerge. **First**, the fine-tuned 0.8B (trained macro-F1 0.468) surpasses every larger base model including the 9B variant (0.431), a model $11\times$ its size, because task-specific supervised fine-tuning compensates for smaller capacity. At 1.5 GB inference VRAM, the fine-tuned 0.8B is feasible on flagship smartphones, enabling on-device deployment. **Second**, fine-tuning erodes zero-shot transfer at scale: the base 4B achieves 0.395 zero-shot, which drops to 0.362 after fine-tuning (-0.033); the base 9B (0.405) is the best zero-shot model overall. Fine-tuning on english, swahili, and ukrainian specialises the weights toward the training distribution, trading away multilingual generalisation. The

4B fine-tuned model (0.543 trained) is the best in-domain result but is not recommended when broad language coverage is required.

D. Zero-Shot Language Transfer

The three zero-shot languages span typological distance. German (Germanic, high-resource) reaches 0.391 with the fine-tuned 0.8B; Brazilian Portuguese (Romance, high-resource) reaches 0.396, the highest zero-shot score, benefiting from typological proximity to the training languages. Yoruba (Niger-Congo, low-resource) reaches only 0.097 with fine-tuned 0.8B and 0.247 with base 9B, despite a $12\times$ parameter increase. This ceiling effect indicates a pretraining data coverage problem: scale and fine-tuning cannot close a gap caused by script under-representation.

E. Chinese: Does Model Heritage Help?

Qwen3.5 is trained by Alibaba with heavy Chinese-language data, so Chinese is a natural test case for whether model heritage translates into downstream emotion-detection advantage. Chinese (Mandarin) is the only CJK language in BRIGHTER, with 400 development examples evaluated as a zero-shot language in our setup.

TABLE III
CHINESE MACRO-F1 (ZERO-SHOT) ACROSS MODEL SIZES. FT = FINE-TUNED ON ENG/SWA/UKR ONLY; CHINESE WAS NEVER IN THE TRAINING SET.
BOLD MARKS THE BEST RESULT.

Size	Base	FT (eng+swa+ukr)	Δ
0.8B	0.332	0.466	+0.134
2B	0.399	0.483	+0.084
4B	0.563	0.555	-0.008
9B	0.531	n/a	n/a

Chinese is a strong zero-shot language: the 0.8B base model ranks 3rd of 28 BRIGHTER languages (0.332) and the 4B ranks 6th (0.563), confirming that Qwen’s pretraining gives Chinese a genuine head start. However, heritage does not exempt it from the general fine-tuning erosion pattern: at 4B, fine-tuning on other languages slightly hurts Chinese (0.563 to 0.555, -0.008). Chinese is also the only top-seven language where 9B does not improve over 4B (0.531 vs. 0.563), suggesting the 4B is particularly well suited to Chinese-script emotion recognition.

At smaller sizes, fine-tuning does help: the 0.8B gains $+0.134$ because the small base model lacks sufficient zero-shot emotion knowledge, and task training on any language teaches the general concept of emotions. Examining per-label shifts at 0.8B, fine-tuning recovers fear (0.000 to 0.235) and surprise (0.175 to 0.468), but collapses disgust (0.409 to 0.105), because the fine-tuning distribution has very few disgust examples. In summary, Qwen’s Chinese heritage provides a meaningful zero-shot advantage, but that advantage is best preserved by using a large base model rather than fully fine-tuning on other languages.

F. LIME Error Analysis

LIME [3] probes which words most influenced each prediction by fitting a local linear surrogate on 500 perturbed inputs. For Qwen3.5, discrete generation outputs are converted to proxy probabilities (0.95 positive / 0.05 negative per label), so attributions are qualitative error signals rather than calibrated confidence estimates. We selected one misclassified example per evaluation language and ran explanations for both the fine-tuned Qwen3.5-0.8B and the mBERT baseline.

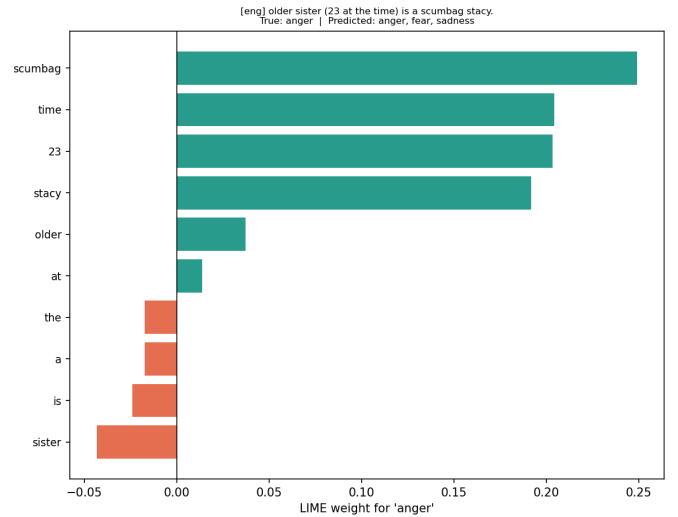


Fig. 2. LIME attributions for the English anger example (Qwen3.5-0.8B fine-tuned, 500 perturbations). Green bars push the prediction toward **anger**; red bars push against it. “scumbag” dominates at $+0.249$; function words carry near-zero weight. Whilst the model correctly predicts anger, it also falsely predicts fear and sadness, hence the classification is labeled as false.

English (Fig. 2): the input “older sister (23 at the time) is a scumbag stacy.” carries true label anger. “scumbag” is the top attribution in both Qwen ($+0.249$) and mBERT ($+0.236$), with function words near zero. Explicit offensive vocabulary produces consistent, architecture-independent LIME signal, confirming that lexically unambiguous sentiment cues transfer reliably across scripts and model families.

Ukrainian (Fig. 3): the text describes going to a concert, raising a toast (“за здоров’я друга”, “to a friend’s health”), and eating cake - a clearly joyful scene. mBERT’s dominant attribution is “здоров” (“healthy”) ($+0.401$), which the model associates with illness and medical contexts rather than the celebratory formula, predicting **sadness** instead of **joy**. Qwen makes the complementary error: it predicts nothing, with attributions an order of magnitude smaller (top weight $+0.013$), indicating uncertainty rather than active mislabelling.

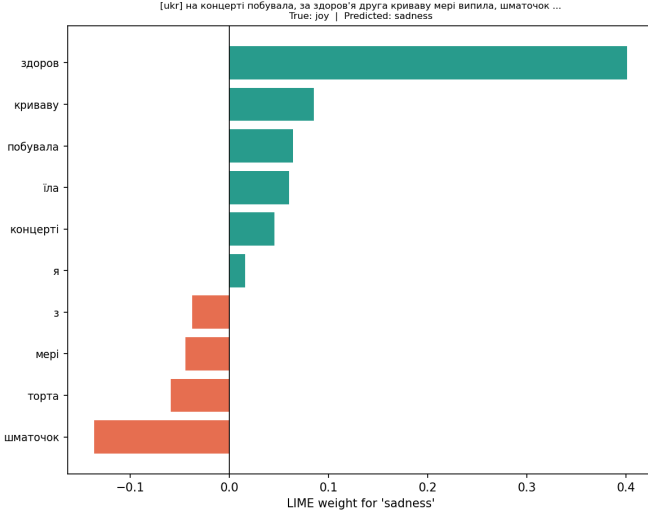


Fig. 3. LIME attributions for the Ukrainian example (mBERT, explained label: **sadness**). True label is **joy**. “здоров” (from the celebratory toast formula “за здоров’я”, “to your health”) receives the highest positive weight (+0.401) but drives the wrong emotion.

IV. DISCUSSION

The results expose a clean trade-off between two deployment objectives. For **maximum performance on known languages with minimal hardware**, fine-tuning a small model is the right strategy: the fine-tuned 0.8B (1.5 GB VRAM) outperforms the base 9B (17.9 GB VRAM) on trained languages, and the fine-tuned 4B (0.543) is the best absolute result in the study.

For **maximum coverage of unknown languages**, fine-tuning is counterproductive at scale: base 4B zero-shot (0.395) beats fine-tuned 4B (0.362), and a large base model is preferable. These objectives cannot be simultaneously optimised with full fine-tuning; lighter-touch adaptation (LoRA or fewer training languages) could preserve more zero-shot breadth while still gaining task knowledge.

A systematic search over hyperparameters, data reweighting, and prompt design found no improvement beyond the noise floor (± 0.008 macro-F1 at $n = 5$ seeds).

Rare-emotion oversampling monotonically degraded performance (-0.023 gentle, -0.038 strong): naive row duplication shifts the prediction prior without adding information, increasing false positives for rare labels without fixing their recall.

A recall-oriented system prompt with explicit cues for hard-to-detect emotions produced a statistically insignificant change (epoch-3 delta -0.005 , Welch $p = 0.50$). These negative results strongly suggest that the only remaining lever at fixed model size is genuinely new data, for example back-translation augmentation for Swahili’s fear-label (0.068 ± 0.070).

Low-resource African languages present a structural challenge that capacity alone cannot solve. Swahili’s 0.273 trained macro-F1 is pulled down by rare-emotion cells (fear F1 = 0.100) that remain unstable across seeds. Yoruba reaches only 0.097 even after fine-tuning; even base 9B achieves only

0.247, suggesting the bottleneck is pretraining coverage, not model size. The two approaches show complementary failures: mBERT outperforms Qwen on Yoruba (0.169 vs. 0.097) because the encoder’s sigmoid outputs are less biased toward abstention.

The Chinese sub-study reinforces the heritage-versus-fine-tuning trade-off. Qwen’s pretraining gives Chinese a strong zero-shot start (4B base ranks 6th of 28 BRIGHTER languages at 0.563), but fully fine-tuning on eng/swa/ukr erodes this advantage at scale (4B FT: 0.555, -0.008). Notably, 4B outperforms 9B on Chinese (0.563 vs. 0.531), the only top-seven language where this inversion occurs, indicating that the 4B is the Chinese sweet spot even without fine-tuning.

V. CONCLUSION

Our results expose a clear trade-off between compact, task-specific performance and broad zero-shot transfer. The central practical finding is that a fine-tuned 0.8B model (1.5 GB at inference) outperforms every larger base model on trained languages, including a model $11\times$ its size, making on-device deployment a realistic option. This gain comes at a cost: full fine-tuning at scale erodes zero-shot transfer, so for multilingual breadth without a training budget, a large base model is preferable to task-specific fine-tuning. The remaining challenge is data: low-resource languages such as Yoruba stay near the performance floor regardless of model size, and information-adding augmentation such as back-translation for rare Swahili emotion categories is the natural next step.

VI. ARTIFACTS

We published our models publically to huggingface [6]. All other artifacts are available in the published github repository [7].

REFERENCES

- [1] Q. Team, “Qwen3 Technical Report.” [Online]. Available: <https://arxiv.org/abs/2505.09388>
- [2] T. Radeva and others, “SemEval-2025 Task 11: Bridging the Gap in Text-Based Emotion Recognition,” in *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*, 2025.
- [3] M. T. Ribeiro, S. Singh, and C. Guestrin, “‘Why Should I Trust You?’: Explaining the Predictions of Any Classifier,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144. doi: 10.1145/2939672.2939778.
- [4] Google BERT, “BERT multilingual base model (cased).” Accessed: Jun. 18, 2026. [Online]. Available: <https://huggingface.co/google-bert/bert-base-multilingual-cased>
- [5] T. Dettmers, M. Lewis, S. Shleifer, and L. Zettlemoyer, “8-bit Optimizers via Block-wise Quantization,” *9th International Conference on Learning Representations, ICLR*, 2022.
- [6] C. G. Niklas Steffen Julian Steffen, “MMD2 – BRIGHTER Multilingual Emotion Classifiers.” Hugging Face, 2026.
- [7] C. G. Niklas Steffen Julian Steffen, “MMD2 – Submission Repository.” Hugging Face, 2026.